

Chapter 9a: Approximating Solutions Applications of Fixed Point Theorems

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Nonlinear Analysis and Fixed Point Theory

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Banach Spaces: Foundational Results

Lemma 9.1

Let T be locally quasi-nonexpansive at $p \in \tilde{\mathcal{F}}(T)$ w.r.t. $\{x_n\}$, and

$$\lim_{n \rightarrow \infty} d(x_n, \tilde{\mathcal{F}}(T)) = 0$$

Then, $\{x_n\}$ is a Cauchy sequence.

Theorem 9.3 (Pathak)

Let $\tilde{\mathcal{F}}(T)$ be a nonempty closed set. Then:

- ① $\lim_{n \rightarrow \infty} d(x_n, \tilde{\mathcal{F}}(T)) = 0$ if $\{x_n\}$ converges to $p \in \tilde{\mathcal{F}}(T)$;
- ② $\{x_n\}$ converges to a point in $\tilde{\mathcal{F}}(T)$ if $\lim_{n \rightarrow \infty} d(x_n, \tilde{\mathcal{F}}(T)) = 0$, T is locally quasi-nonexpansive at $p \in \tilde{\mathcal{F}}(T)$ w.r.t. $\{x_n\}$, and X is complete.

These results provide convergence guarantees in Banach spaces.

Theorem 9.4: Geometry and Approximation

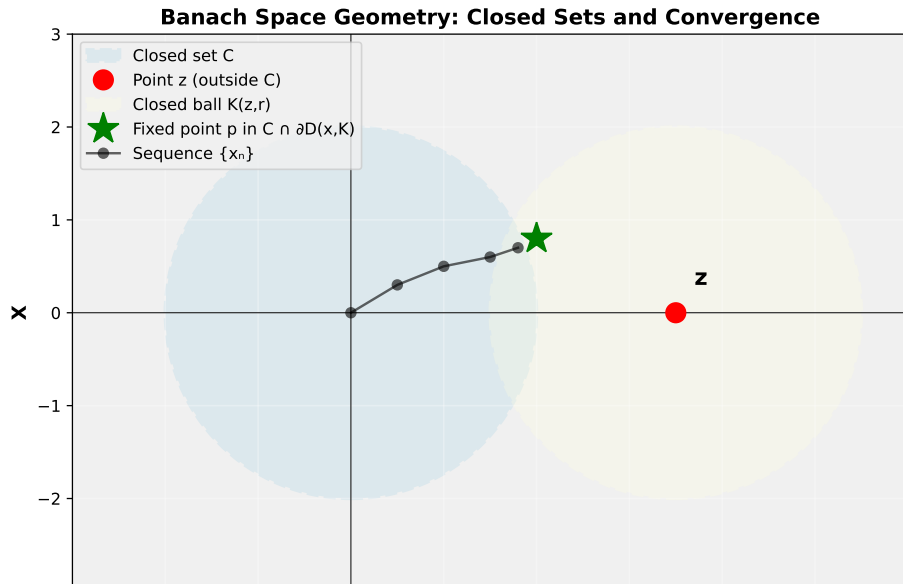
Theorem 9.4 (Pathak)

Let C be a closed subset of Banach space X , let $z \in X \setminus C$, and let $K = K(z, r)$ be a closed ball with radius $r = d(z, C) = R$. Let x be arbitrary in C , let $\{x_n\}$ be a sequence in C converging to x , and let $T : C \rightarrow X$ be continuous with $T(x_n) \in C \cap \mathcal{D}(x, K)$ for each $n \in \omega$. Then:

- ① $\lim_{n \rightarrow \infty} d(x_n, \tilde{F}(T)) = 0$ if $\{x_n\}$ converges to a point $p \in F(T)$;
- ② $\{x_n\}$ converges to a point in $\tilde{F}(T)$ if $\lim_{n \rightarrow \infty} d(x_n, \tilde{F}(T)) = 0$ and T is locally quasi-nonexpansive at $p \in \tilde{F}(T)$ w.r.t. $\{x_n\}$.

This geometric approach provides a bridge between fixed point theory and approximation in Banach spaces.

Geometric Visualization



Linear System and Fixed Point Formulation

Consider the system of n linear equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases}$$

Rewrite as: $\mathbf{x} = \mathbf{x} - A\mathbf{x} + \mathbf{b}$ where $A = (a_{ij})_{n \times n}$ and $\mathbf{b} = (b_1, \dots, b_n)^T$.

Key Idea: Apply Banach contraction theorem with $T(\mathbf{x}) = \mathbf{x} - A\mathbf{x} + \mathbf{b}$ to find unique solution $\mathbf{x}^*$.

Convergence Condition: Spectral radius $\rho(I - A) < 1$ ensures contraction.

Linear System: Iterative Algorithm

Jacobi Iteration for Linear Systems

Given: Matrix A and vector $\mathbf{b}$, with $a_{ii} \neq 0$.

Fixed point iteration:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right), \quad i = 1, 2, \dots, n$$

Convergence: If A is diagonally dominant, the iteration converges.

Example: Consider the system

$$\begin{pmatrix} 4 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Apply Jacobi iterations starting from $\mathbf{x}^{(0)} = (0, 0, 0)^T$ to obtain the solution.

Nonlinear Matrix Equation Problem

Problem Setup

Consider the matrix equation:

$$X = Q + \sum_{i=1}^m A_i X^{\delta_i} A_i^*, \quad 0 < \delta_i < 1$$

where:

- Q is an $n \times n$ positive semidefinite matrix
- A_i are nonsingular $n \times n$ matrices
- X is the positive definite solution

Applications: Control theory, stability analysis, and system identification.

Thompson Metric: On cone $P(N)$ of Hermitian positive definite matrices:

$$d(A, B) = \max\{\log M(A/B), \log M(B/A)\}$$

where $M(A/B) = \lambda_{\max}(B^{-1/2}AB^{-1/2})$ is the maximal eigenvalue.

Theorem 9.8: Convergence to Unique Solution

Theorem 9.8

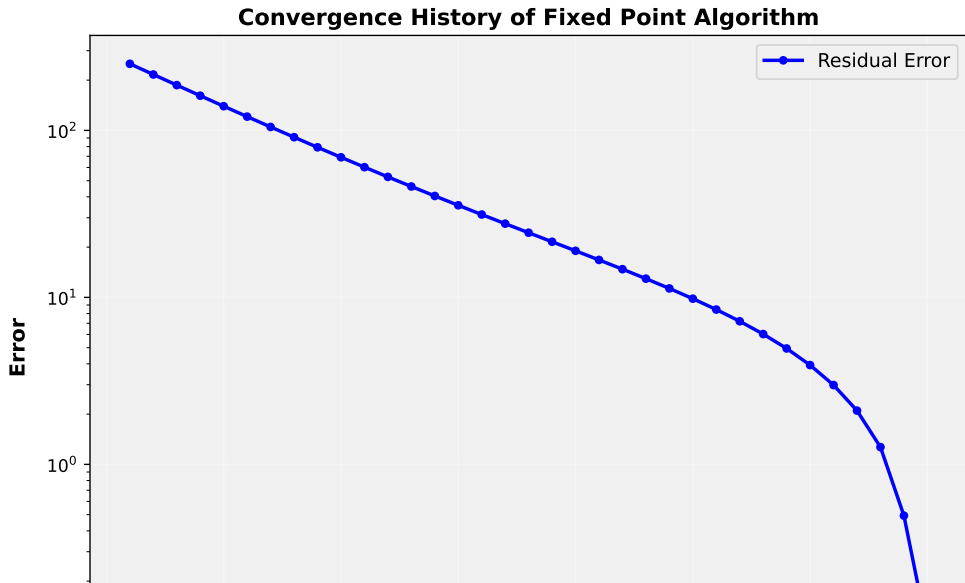
Suppose $\lambda = \max\{|\alpha_1|, |\alpha_2|, \dots, |\alpha_{r+1}|\} \in (0, 1)$ where α_i are real parameters. Then:

- 1 The equation $X = Q + A_1^* X^{\alpha_1} A_1 + A_2^* X^{\alpha_2} A_2 + \dots + A_{r+1}^* X^{\alpha_{r+1}} A_{r+1}$ has a **unique equilibrium point** $U \in P(N)$ satisfying

$$U = Q + A_1^* U^{\alpha_1} A_1 + A_2^* U^{\alpha_2} A_2 + \dots + A_{r+1}^* U^{\alpha_{r+1}} A_{r+1}$$

- 2 The iterative sequence $\{X_n\}$ defined by the recursion **converges to the unique solution** of the matrix equation.

Numerical Example: Matrix Convergence



Control Theory: Problem Formulation

Optimal Control Problem

Consider the ordinary differential equation (ODE):

$$\dot{x}(s) = f(x(s), \alpha(s)), \quad x(t) = x_0$$

where $\alpha(\cdot) \in \mathcal{A}$ is an admissible control.

Find optimal control $\alpha^*(\cdot)$ that minimizes the cost functional:

$$C_{x_0}[\alpha(\cdot)] := \int_t^T h(x(s), \alpha(s)) ds + g(x(T))$$

Key Components:

- Running cost: $h(\mathbf{x}, \alpha)$ per unit time
- Terminal cost: $g(\mathbf{x}(T))$ at final time
- Value function: $u(x, t) := \inf_{\alpha \in \mathcal{A}} C_{x,t}[\alpha]$

Control Theory: Value Function and Hamilton-Jacobi PDE

Value Function Properties

The value function $u(x, t)$ solves Hamilton-Jacobi PDE:

$$u_t(x, t) + \inf_{\alpha \in \mathcal{A}} \{h(x, \alpha) + \nabla u \cdot f(x, \alpha)\} = 0$$

with terminal condition $u(x, T) = g(x)$.

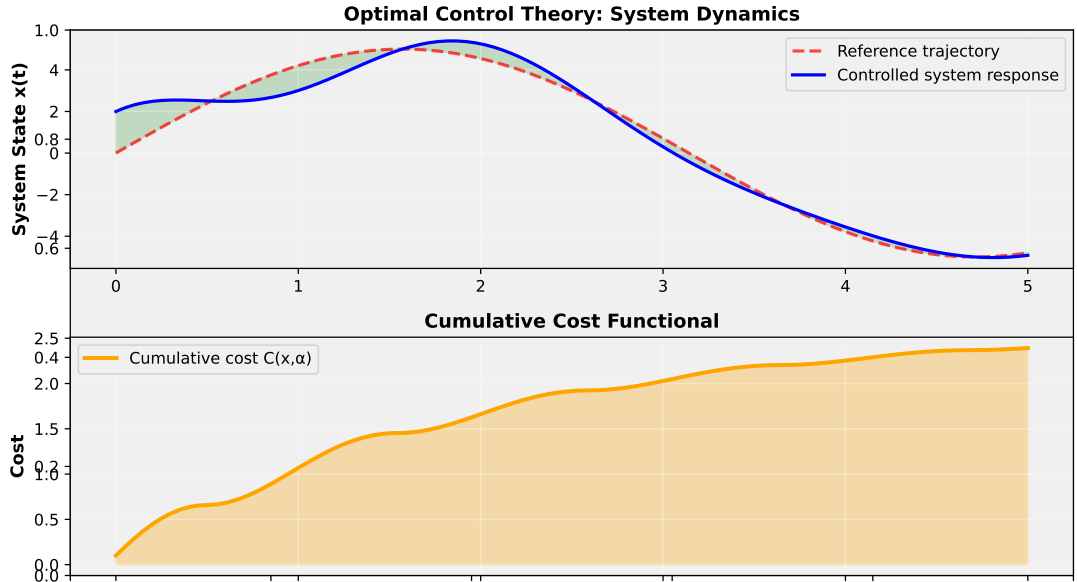
Optimality Conditions

For each $\xi > 0$ small with $t + \xi \leq T$:

$$u(x, t) := \inf_{\alpha \in \mathcal{A}} \left\{ \int_t^{t+\xi} h(x(s), \alpha(s)) ds + u(x(t + \xi), t + \xi) \right\}$$

Application: Dynamic programming approach using fixed point methods on operator equations.

Control System Visualization



Variational Inequalities in Stochastic Control

Two-Obstacle Variational Inequality

Find function u such that:

$$\max\{\min\{Lu - f, u - \phi\}, u - \mu\} = 0 \quad \text{on } \Omega$$

with boundary condition $u = 0$ on $\partial\Omega$.

Here:

- $L = -a_{ij}(x)\frac{\partial^2}{\partial x_i \partial x_j} + b_i(x)\frac{\partial}{\partial x_i} + c(x)I_N$ is elliptic operator
- $\phi < \mu$ are obstacle functions
- Solution must lie in feasible region $[\mu, \phi]$

Interpretation: Two competing controls; the solution reflects optimal play.

M-Matrix and Discrete Variational Inequality

Definition

M-matrix A square matrix A is an M-matrix if and only if every off-diagonal entry is nonpositive.

Discrete Two-Obstacle Problem

For discrete case with B-matrix $B = I_N - A$:

$$\max\{\min\{A(x - A^* d(x, Tx)) - f, x - A^* d(x, Tx) - \phi\}, x - A^* d(x, Tx) - \mu\} = 0$$

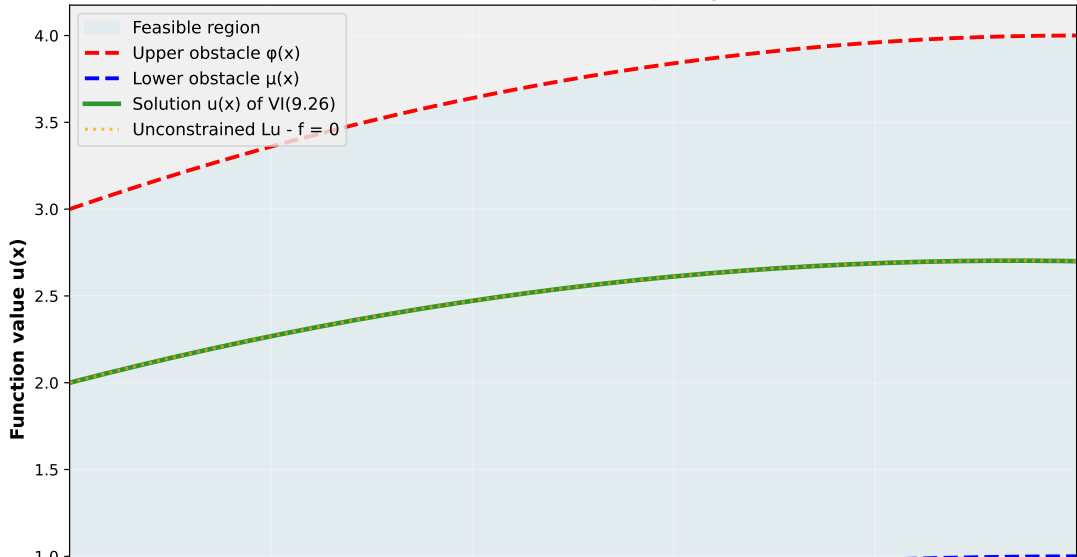
where T is defined implicitly by the min-max conditions.

Theorem 9.9

Under assumptions (9.27) and (9.28) on matrices, a solution to equation (9.31) exists.

Variational Inequality Visualization

Two-Obstacle Variational Inequality Problem



Two-Person Zero-Sum Game

Definition

Nash Equilibrium A pair of strategies $(x^*, y^*) \in \Delta^m \times \Delta^n$ (simplices) is a Nash equilibrium if:

$$\begin{aligned} e_1(x, y^*) &\leq e_1(x^*, y^*), & \text{for all } x \in \Delta^m \\ e_2(x^*, y) &\leq e_2(x^*, y^*), & \text{for all } y \in \Delta^n \end{aligned}$$

where $e_i(\mathbf{x}, \mathbf{y})$ are expected payoff functions for players 1 and 2.

Key Properties:

- (x^*, y^*) is a fixed point of the best-response mapping
- Neither player can unilaterally improve payoff by changing strategy
- Probability-based mixed strategies determine average outcomes

Expected Payoffs: For payoff matrix $P = (p_{ij})$:

$$e_1(\mathbf{x}, \mathbf{y}) = \mathbf{x}P\mathbf{y}^T = \sum_{i=1}^m \sum_{j=1}^n x_i y_j p_{ij}$$

Theorem 9.10: Existence of Nash Equilibrium

Theorem 9.10 (Fundamental Theorem of Game Theory)

For a two-player zero-sum game, there exists a Nash equilibrium.

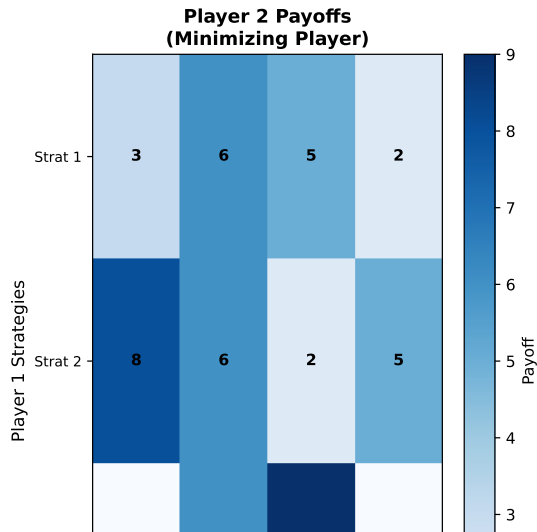
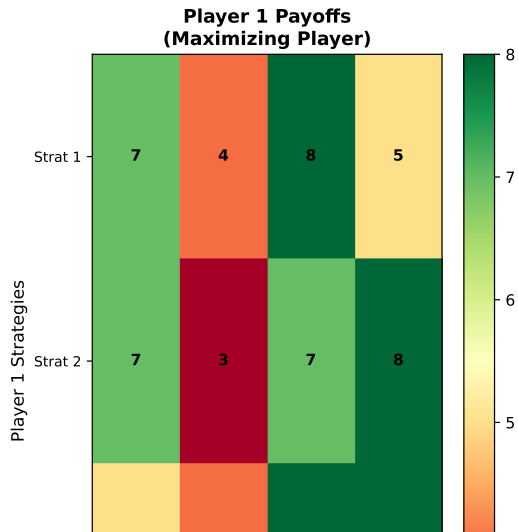
Proof Sketch:

- 1 The space of all pairs $(\mathbf{x}, \mathbf{y})$ is homeomorphic to $(m + n)$ -dimensional disc $\Delta^m \times \Delta^n$
- 2 Define continuous mapping $g : \Delta^m \times \Delta^n \rightarrow \Delta^m \times \Delta^n$ where fixed point represents Nash equilibrium
- 3 Apply Brouwer's fixed point theorem to guarantee existence
- 4 The fixed point satisfies all equilibrium conditions

Conclusion: Every finite two-person zero-sum game has at least one Nash equilibrium (possibly in mixed strategies).

Nash Equilibrium Payoff Analysis

Two-Person Zero-Sum Game: Nash Equilibrium Analysis



9.4.1: Picard-Lindelof Existence Theorem

Theorem 9.11 (Picard-Lindelof)

Let Ω denote an open set in $\mathbb{R}^2$, $(t_0, x_0) \in \Omega$. Let $f(t, x)$ be real-valued, continuous in Ω , and satisfy the Lipschitz condition

$$|f(t, x_1) - f(t, x_2)| \leq M|x_1 - x_2|, \quad (t, x_1), (t, x_2) \in \Omega$$

Then there exists $\tau > 0$ and a unique function $\varphi(t)$ continuous and differentiable in $[t_0 - \tau, t_0 + \tau]$ such that:

- 1 $\varphi(t_0) = x_0$
- 2 $\dot{\varphi}(t) = f(t, \varphi(t))$ for $t \in [t_0 - \tau, t_0 + \tau]$

Proof Strategy: Reduce ODE to Hammerstein integral equation, then apply Banach contraction theorem on appropriate function space.

Picard-Lindelof: Integral Formulation

Reduction to Integral Equation

The ODE $\dot{x}(t) = f(t, x(t))$ with initial condition $x(t_0) = x_0$ is equivalent to:

$$\varphi(t) = x_0 + \int_{t_0}^t f(s, \varphi(s)) ds$$

Contraction Mapping on $E = C[t_0 - \tau, t_0 + \tau]$

Define metric space $(E, \|\cdot\|)$ where $\|u\| = \max_{t \in [t_0 - \tau, t_0 + \tau]} |u(t)|$.

Define $\Psi : E \rightarrow E$ by

$$\Psi(u)(t) = x_0 + \int_{t_0}^t f(s, u(s)) ds$$

For $\tau m < 1$ (where m is upper bound of $|f|$), Ψ is contraction with unique fixed point.

Key: Fixed point of Ψ is the unique solution to the ODE.

Differential Equation Example

Example: Linear First-Order ODE

Solve $\dot{x}(t) = \lambda x(t)$ with $x(0) = x_0$ using fixed point method.

The integral equation becomes:

$$\varphi(t) = x_0 + \int_0^t \lambda \varphi(s) ds$$

Successive approximations:

$$\varphi_0(t) = x_0$$

$$\varphi_1(t) = x_0 + \int_0^t \lambda x_0 ds = x_0(1 + \lambda t)$$

$$\varphi_2(t) = x_0 + \int_0^t \lambda x_0(1 + \lambda s) ds = x_0 \left(1 + \lambda t + \frac{(\lambda t)^2}{2!} \right)$$

$$\varphi_n(t) \rightarrow x_0 e^{\lambda t} \quad \text{as } n \rightarrow \infty$$

This recovers the classical solution!

Theorem 9.12: Convergence of Successive Approximations

Theorem 9.12

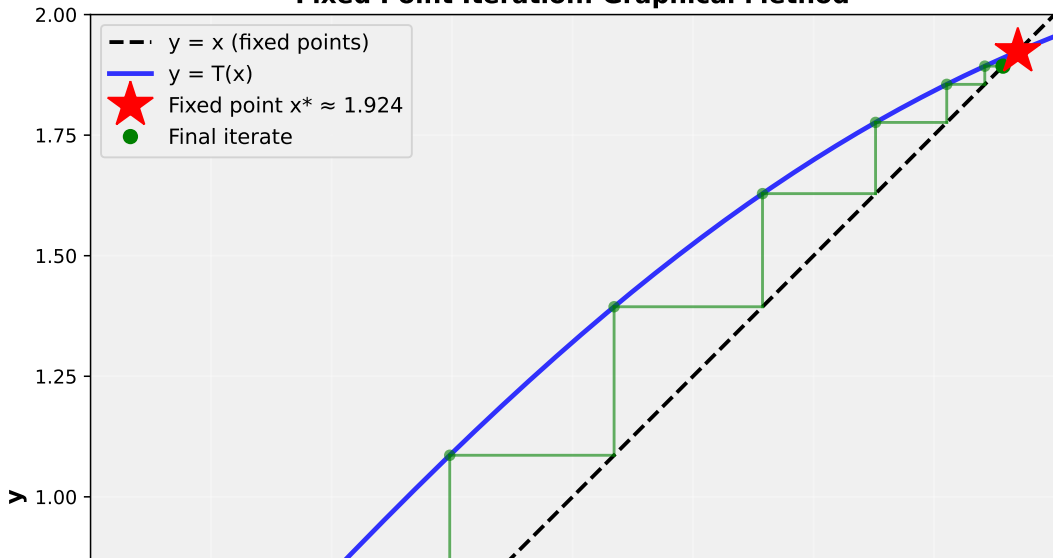
If φ_n is the solution of the differential equation (9.39) with initial condition $\varphi_n(t_0) = x_0$ and $\{x_n\}$ is a real sequence converging to x_0 , then $\{\varphi_n\}$ converges to the solution φ of (9.39) with $\varphi(t_0) = x_0$ in E .

Proof: The Picard iterates form a Cauchy sequence in the complete metric space $C[t_0 - \tau, t_0 + \tau]$, converging to the unique fixed point of the contraction mapping.

Practical Significance: Allows numerical solution via iterative computation of successive approximations.

Fixed Point Iteration Method

Fixed Point Iteration: Graphical Method



Key Takeaways

Theory Highlights

- Banach contraction theorem extends to approximation problems
- Convergence depends on metric structure and contraction ratios
- Applications span diverse fields: linear/nonlinear systems, control, game theory

Practical Applications

- Solving large-scale matrix equations
- Optimal control and dynamic programming
- Game-theoretic equilibrium computation
- Existence proofs for PDEs and ODEs

Unifying Principle

All applications use the same core technique: reduce the problem to finding a fixed point of a contraction mapping, then invoke the Banach contraction theorem to guarantee existence, uniqueness, and convergence.

Convergence Rates and Computational Complexity

Linear Convergence

$$d(x_n, x^*) \leq \lambda^n d(x_0, x^*)$$

Convergence factor $\lambda = \max\{|\alpha_i|\}$:

- $\lambda < 0.5$: Fast convergence
- $0.5 < \lambda < 1$: Moderate convergence
- λ close to 1: Slow convergence

Practical Considerations

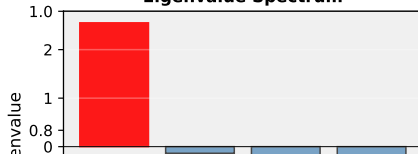
- Iteration count n^* for tolerance ϵ :

$$n^* \approx \frac{\ln(\epsilon/d_0)}{\ln \lambda}$$

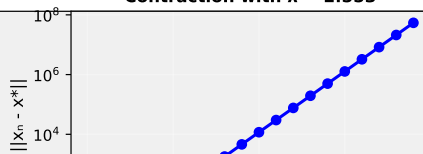
- Trade-off: simple iteration vs. complicated fixed point operator
- Preconditioners improve contraction ratio

Matrix Properties and Convergence Analysis

Eigenvalue Spectrum



Contraction with $\lambda = 2.553$



Python Implementation Example

```
import numpy as np

def fixed_point_iteration(T, x0, tol=1e-6, max_iter=1000):
    """
    Fixed point iteration:  $x_{n+1} = T(x_n)$ 
    """
    x = x0.copy()
    errors = []

    for n in range(max_iter):
        x_new = T(x)
        error = np.linalg.norm(x_new - x)
        errors.append(error)

        if error < tol:
            print(f"Converged in {n} iterations")
            return x_new, errors

    x = x_new

    print(f"Max iterations reached: {max_iter}")
    return x, errors

# Example: Solve linear system  $Ax = b$  via  $T(x) = x - Ax + b$ 
A = np.array([[2, 0.5], [0.5, 2]])
b = np.array([1, 1])
x0 = np.zeros(2)

T = lambda x: x - A @ x + b
solution, errors = fixed_point_iteration(T, x0)
print(f"Solution: {solution}")
```

Future Extensions and Research Directions

- **Nonlinear Operators:** Extension to monotone operators and variational inequalities
- **Higher-Dimensional Systems:** Scaling and parallelization for large-scale problems
- **Adaptive Methods:** Dynamic contraction rates based on computed residuals
- **Machine Learning:** Neural network approximations of fixed point mappings
- **Hybrid Methods:** Combining fixed point iterations with gradient-based optimization

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- Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales
- Caristi, J. (1976). Fixed point theorems for mappings satisfying inwardness conditions